Big Data Project

From Data to Knowledge: An Investigation to Share

Objective

Choose a **topic that interests you**, find a relevant dataset, and carry out an analysis to answer **a specific question**. Your task does not end with producing graphs: you must draw conclusions from your analysis and then **share your findings with the class**.

Organisation

The project can be completed **individually or in pairs**. You are **free to choose** your topic and dataset. They must allow for a meaningful analysis and conclusions supported by the data.

Deliverable 1 — Written Summary (2 pages maximum)

Your document should make it possible to understand your work without reading all your code:

- **The research question** and why it is interesting;
- **The data**: source, content, and any limitations;
- **The approach**: cleaning, transformations, and methodological choices;
- **The results**: a few relevant figures, interpreted and justified;
- **The conclusion**: an answer to your question, including the limitations of the analysis.

Prioritise clear reasoning over an accumulation of numbers or graphs.

Deliverable 2 — Interactive Quiz (5 minutes maximum)

Present **5 multiple-choice questions** to the class, each with **5 possible answers** (only one correct). Your classmates will participate, for example by raising their hands.

For each question:

- Give the class time to think before revealing the answer;
- Show a result, a number, or a figure from your analysis;
- Briefly explain why the correct answer is supported by the data.

Aim to share **five interesting, surprising, or counterintuitive findings**. This is not a general-knowledge quiz: the answers must come from **your own analysis**.

Assessment Criteria

The relevance of the question, the coherence of the method, the quality of the visualisations, the rigour of the interpretation, critical thinking, and the ability to communicate results clearly.

The goal is not to produce as many results as possible, but to select those worth sharing.